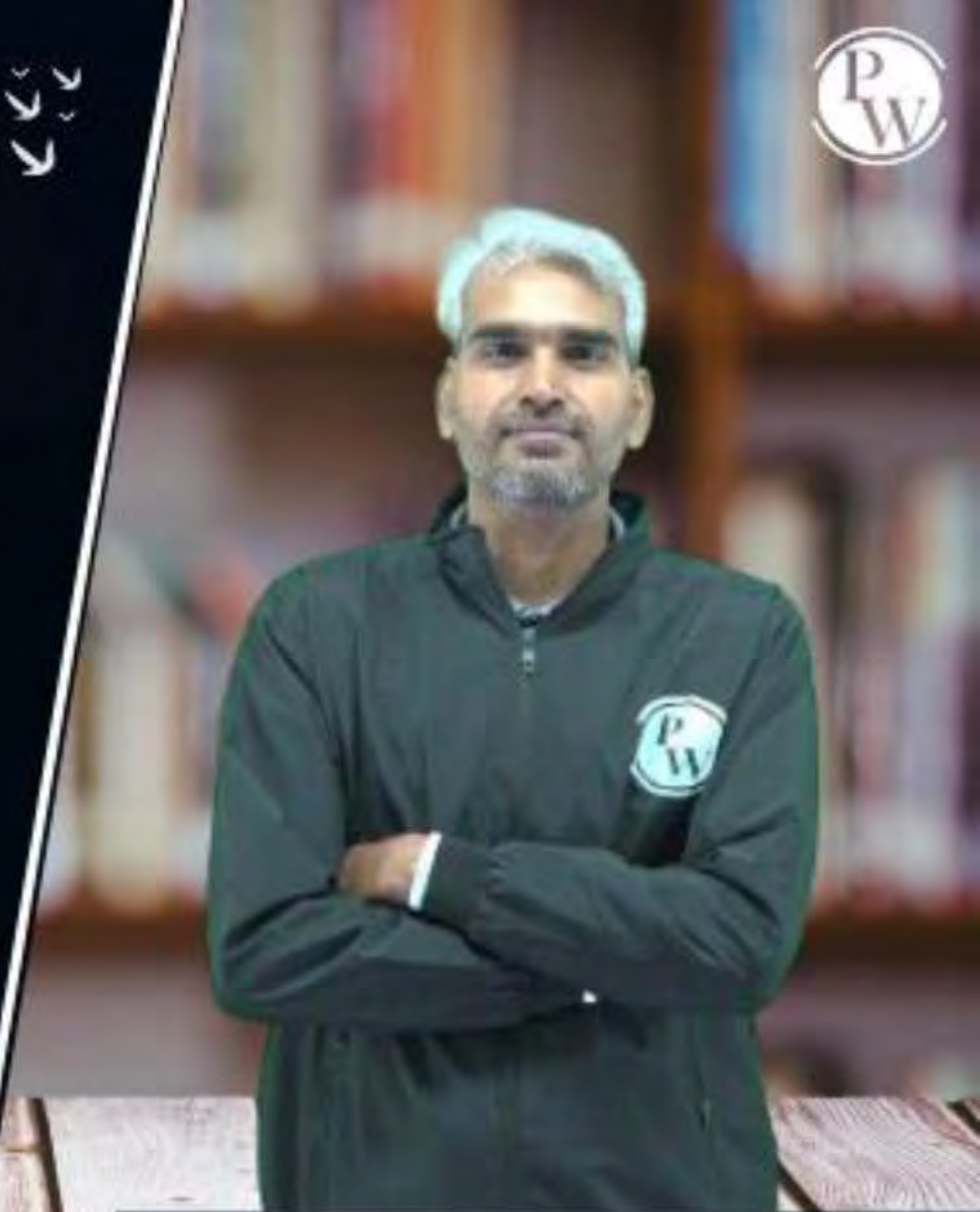
Computer Science & IT

Discrete Mathematics

Combinatorics

Lecture No. 01



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Recap of Previous Lecture

Topic — Practice questions of predicate logic

Topic — Tautology w.r.t. predicate formula

Slide

Topics to be Covered

- **Topic** Introduction to recurrence relation
- **Topic** Formation of recurrence relation
- **Topic** Solution of recurrence relation using substitution method

Slide

[MSQ]

H.W.

#Q. What is the correct translation of the following statement into mathematical logic?

"some real numbers are rational"

A $\exists x\,(real(x) \vee rational(x))$

B $\exists x\,(real(x) \wedge rational(x))$

C $\forall x\,(real(x) \rightarrow rational(x))$

D $\forall x\,(rational(x) \rightarrow real(x))$

[MSQ]

H.W.

#Q. What is the logical translation of the following statement?

$\sim \exists_x \{ F(x) \wedge P(x) \}$

Not at least one of my friend is perfect

"None of my friends are perfect"

All my friends are imperfect

A $\exists x\,(F(x) \wedge \sim P(x))$

B $\exists x\,(\sim F(x) \wedge \sim P(x))$

$\sim\sim\left(\forall_x \{ F(x) \longrightarrow \sim P(x) \} \right)$

C $\exists x\,(\sim F(x) \wedge P(x))$

$\sim \left[\exists_x \{ \sim (F(x) \rightarrow \sim P(x)) \} \right]$

D $\sim (\exists x\,(F(x) \wedge P(x)))$ ✓

$\sim \exists_x \{ \sim (\sim F(x) \vee \sim P(x)) \}$

$\sim \exists_x \{ F(x) \wedge P(x) \}$

H.W. [MSQ]

#Q. The CORRECT formula of the sentence "not all rainy days are cold"

A. $\forall d\,(rainy(d) \wedge \sim cold(d))$

B. $\exists d\,(\sim rainy(d) \rightarrow cold(d))$

C. $\forall d\,(\sim rainy(d) \rightarrow cold(d))$

D. $\exists d\,(rainy(d) \wedge \sim cold(d))$ ✓

$\sim [\forall_d (rainy(d) \rightarrow Cold(d))]$

$\exists_d \{\sim(rainy(d) \rightarrow Cold(d))\}$

$\exists_d \{\sim(\sim rainy(d) \vee Cold(d))\}$

$\exists_d \{rainy(d) \wedge \sim Cold(d)\}$

H.W. **[MSQ]**

#Q. Which one of the first order predicate calculus statements given below correctly expresses the following English statements? "Tigers and lions attack if they are hungry or threatened".

All tigers & lions if hungry or threatened then Attack

A. ~~$\forall x[(tiger(x) \land lion(x)) \to ((hungry(x) \lor threatened(x)) \to attacks(x))]$~~

B. $\forall x[(tiger(x) \lor lion(x)) \to ((hungry(x) \lor threatened(x)) \land attacks(x))]$

C. ~~$\forall x[(tiger(x) \land lion(x)) \to (attacks(x) \to (hungry(x) \lor threatened(x)))]$~~

D. $\forall x[(tiger(x) \lor lion(x)) \to ((hungry(x) \lor threatened(x)) \to attacks(x))]$ ✓

[MSQ]

$\{P_1, P_2\}$ logically implies $Q \equiv Q$ follow from $\{P_1, P_2\}$

#Q. Which of the following argument ~~is~~ are valid?

H.W

A. $\forall x\,(P(x))$ follows from the premises $\exists x(P(x) \wedge Q(x))$

B. $\exists x(P(x) \wedge Q(x))$ follows from the premises $\exists x(P(x)) \wedge \exists x(Q(x))$

$\{\exists x\{P(x)\} \wedge \exists x\{Q(x)\}\} \xrightarrow[\text{implies}]{\text{logically}} \exists x\{P(x) \wedge Q(x)\}$

C. $\exists x(Q(x))$ follows from the premises $\{\forall x(P(x) \Rightarrow Q(x)), \exists y(Q(y))\}$

$\exists x\{P(x)\}$ will not follow from same Premises

D. $\sim P(a, b)$ follows from the premises $\{\forall x\ \forall y\{P(x, y) \to W(x, y)\}, \sim W(a, b)\}$

option d

$\forall x \forall y \{ P(x,y) \rightarrow W(x,y) \}$, $\sim W(a,b)$

$\equiv$

$P(l,m) \rightarrow W(l,m)$ will be true for all l & m from domain

$\therefore P(a,b) \rightarrow W(a,b)$ will also be true

M.T.

$\therefore \sim P(a,b)$ { $\therefore$ Valid }

[MSQ]

H.W.

#Q. Check whether the following argument is true or false?

$\forall x\{F(x) \rightarrow \sim S(x)\}$ follows from the premises

$\{\underbrace{\exists x\big(F(x) \wedge S(x)\big)}_{P} \rightarrow \underbrace{\forall y(M(y) \rightarrow W(y))}_{Q},\ \exists y\big(M(y) \wedge \sim W(y)\big)\}$

$P \longrightarrow Q$

$\equiv \sim\big(\sim\big(\exists_y (M(y) \wedge \sim W(y))\big)\big)$

$\sim \forall_y (\sim M(y) \vee W(y))$

$\underbrace{\sim \underbrace{\forall_y \{M(y) \longrightarrow W(y)\}}_{Q}}_{\sim Q}$

M.T.

$\therefore \sim P$

i.e. $\sim[\exists x(F(x) \wedge S(x))]$ is true

$\forall x(\sim F(x) \vee \sim S(x))$ is true

Argument is valid.

$$\boxed{\text{Combinatorics}}$$

↓

Recurrence Relation

Topic : Recurrence Relation

Consider a sequence of real numbers $\{a_0, a_1, a_2, a_3, \ldots\}$ a formula that relates a_n with one or more of the preceding terms $\{a_{n-1}, a_{n-2}, \ldots\}$ is called a recurrence relation.

e.g.: $a_n = 3 \cdot a_{n-1} - 2 \cdot a_{n-2}$

$a_n = 2 \cdot a_{n-1} + 5 \cdot a_{n-2} + 2n$

$a_n = f(a_{n-1}, a_{n-2}, \ldots, a_{n-k}, n)$

Topic : Examples of recurrence relation

Fibonacci Series:

Where, $F_0 = 1$ & $F_1 = 1$

$$F_n = F_{n-1} + F_{n-2} \quad , \Rightarrow 1, 1, 2, 3, 5, \ldots$$

Topic : Formation of recurrence relation

Q.1:→ Let a_n represents the number of ways a person can climb a flight of n-steps while person is allowed to skip $\boxed{\text{at most one step at a time}}$, then

Find recurrence relation for a_n

In the first step there are two possibilities

Case 1

Person does not skip any step, and then remaining steps will be (n-1)

or

Case 2

Person skip exactly one step, and then remaining steps will be (n-2)

$$(1 * a_{n-1}) + (1 * a_{n-2})$$

$\therefore$ Recurrence Relation will be

$$\boxed{a_n = a_{n-1} + a_{n-2},}\ n \geq 3$$

$$a_1 = 1$$

$$a_2 = 2$$

Topic : Formation of recurrence relation

Q.2:- Let a_n represents the number of ways a person can climb a flight of n-steps while person is allowed to $\boxed{\text{skip at most two steps}}$ at a time, then

Find recurrence relation for a_n

In the first move, there are three cases

Case ① does not skip any step $\therefore$ Remaining steps $= (n-1)$ → $1 * a_{n-1}$

or

Case ② Skip Exactly '1' step $\therefore$ Remaining steps $= (n-2)$ → $1 * a_{n-2}$

or

Case ③ Skip Exactly '2' steps $\therefore$ Remaining steps $= (n-3)$ → $1 * a_{n-3}$

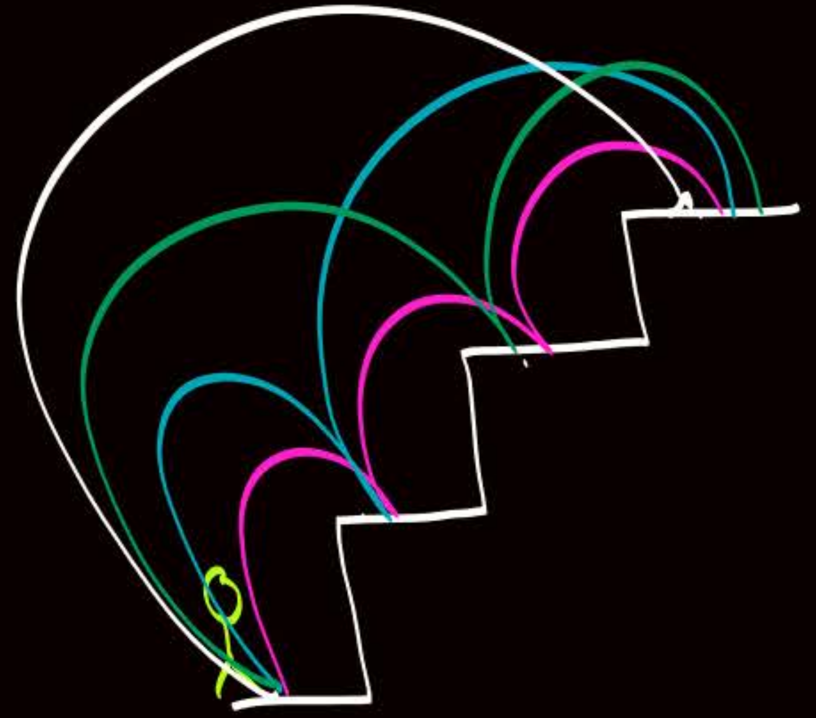
$$1 * a_{n-1} + 1 * a_{n-2} + 1 * a_{n-3}$$

$$\therefore \boxed{a_n = a_{n-1} + a_{n-2} + a_{n-3}}, n \geq 4$$

$a_1 = 1$

$a_2 = 2$

$a_3 = 4$



Topic : Formation of recurrence relation

Q.3:- Let a_n represents the number of ways to arrange a pile of n-chips using Red, Green, Blue, White and Gold colour chips such that no two gold colour chips are together, then

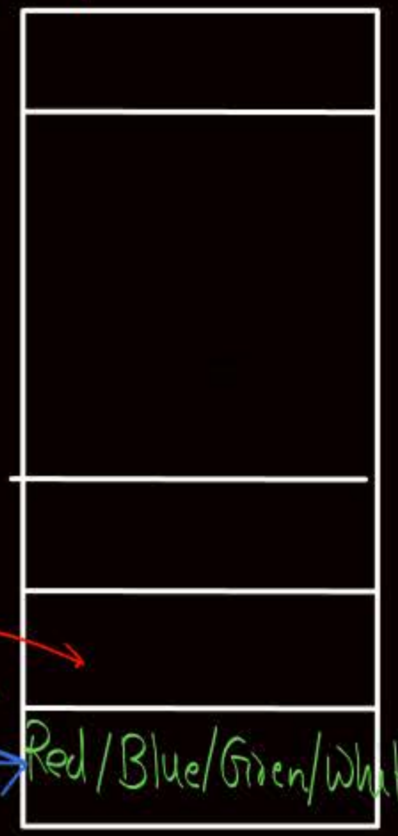
Find recurrence relation for a_n

There are two cases

Case 1
When first chip is not of gold color

Remaining pile of $(n-1)$ chip can be arranged in (a_{n-1}) ways

No restriction

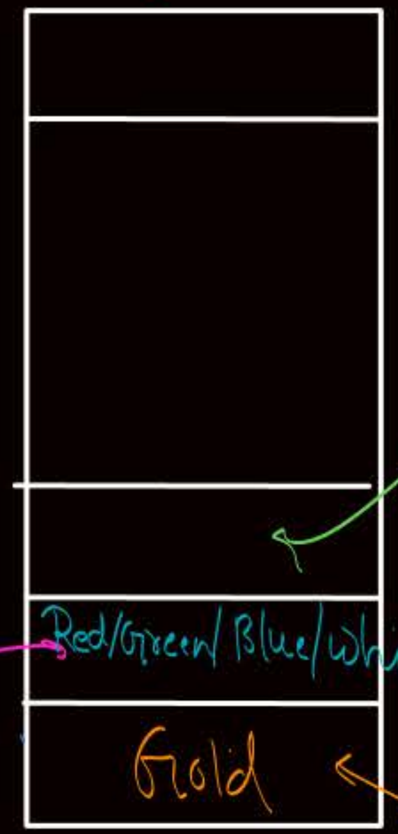


1st chip can be of non-gold color in '4' ways

or

Case 2
When 1st chip is of gold color

No restriction on 3rd chip or Remaining pile of $(n-2)$ chip can be arranged in (a_{n-2}) ways



There is restriction on the color of 2nd chip, that is it must be of non-gold color

1st chip can be gold in '1' way

2nd chip can be non-gold in '4' ways

Recurrence Relation

No. of ways w.r.t. Case 1

No. of ways w.r.t. Case 2

$$a_n = (4 * a_{n-1}) + (1 * 4 * a_{n-2})$$

$$\boxed{a_n = 4 \cdot a_{n-1} + 4 \cdot a_{n-2}}, \; n \geq 3$$

$$a_1 = 5$$

$$a_2 = (1*4 + 4*5) \quad \underline{or} \quad (5^2 - 1)$$

$$= 4 + 20 \qquad = 24$$

$$= 24$$

Topic : Formation of recurrence relation

Q.4:- Let a_n represents the number of n-digit binary sequences of '0' and '1' with no consecutive zeros, then

Find recurrence relation for a_n

There are two Cases

Case ①

When 1^{st} bit is not '0'

'n' bit

1			. . . . . .	

Not '0' only in '1' ways

No restriction on 2^{nd} bit

$\therefore$ for remaining $(n-1)$ bits we have a_{n-1} strings

(Or)

Case ②

When 1^{st} bit is '0'

n bit

0	1		. . . . . .	

1^{st} bit is '0' in '1' ways

restriction on 2^{nd} bit. {i.e it must be non-zero}

$\therefore$ Only '1' way for 2^{nd} bit

No restriction on 3^{rd} bit

$\therefore$ for remaining $(n-2)$ bits, a_{n-2} strings

$$a_n = 1 * a_{n-1} + 1 * 1 * a_{n-2}$$

$$\boxed{a_n = a_{n-1} + a_{n-2}}$$

$a_1 = 2$

$a_2 = 3$

Topic : Formation of recurrence relation

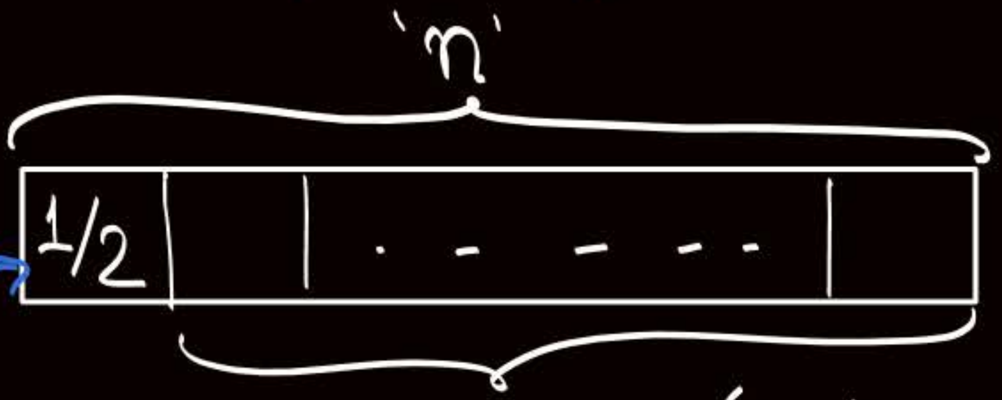
Q.5 : Let a_n represents the number of n-digit ternary sequences of 0, 1, and 2 with even number of zeros in it , then

Find recurrence relation for a_n

There are two cases

Case ①

When 1^{st} digit is not "Zero"



1^{st} digit can be non-Zero in '2' ways

in remaining $(n-1)$ position we need Even no. of Zeros

⇓

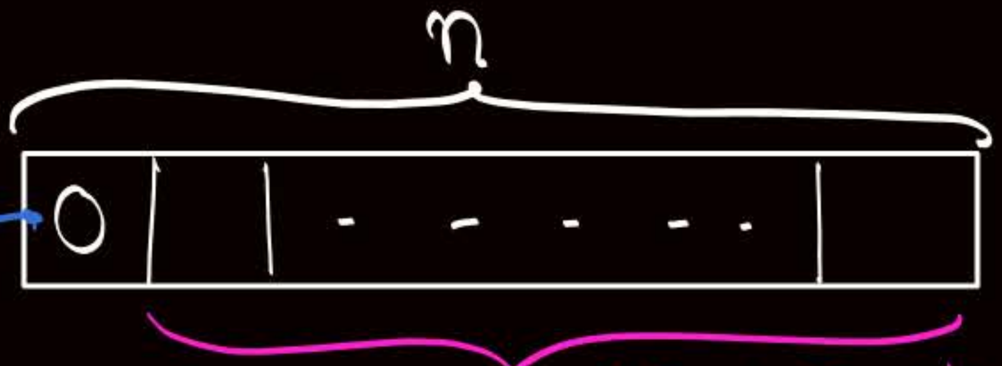
No. of $(n-1)$ digit ternary sequences with Even no. of Zeros can be given by

$$(a_{n-1})$$

(Or)

Case ②

When 1^{st} digit is '0'



1^{st} digit can be '0' in '1' way

in remaining $(n-1)$ position we need odd no. of 0's

⇓

We are interested in no. of $(n-1)$ digit ternary sequences with odd no. of 0's.

⇓

Total no. of $(n-1)$ digit ternary sequences — No. of $(n-1)$ digit ternary sequences with Even no. of 0's

$$(3^{n-1}) - (a_{n-1})$$

$\therefore$ Recurrence Rel^n is

$$a_n = (2 * a_{n-1}) + (1 * (3^{n-1} - a_{n-1}))$$

$$= 2 \cdot a_{n-1} + 3^{n-1} - a_{n-1}$$

$$\boxed{a_n = a_{n-1} + 3^{n-1}} \quad n \geq 2 \qquad \boxed{a_1 = 2}$$

Topic : Solution of recurrence relation

A recurrence relation is a function of the form

$$a_n = f(a_{n-1}, a_{n-2}, \ldots a_{n-k}, n)$$

Representation of a_n as pure function of n {i.e. a function that does not contain any term of type a_{n-i}} is called Solution of recurrence relation

Topic : Solution of recurrence relation

1) $\boxed{\text{Substitution method}}$

2) Method of characteristic roots

3) Method of undetermined coefficients

4) Using generating functions

Topic : Substitution method

In this method we use recurrence relation repetitively for $n=0,1,2,\ldots\ldots..$, then we solve the expression to obtain the solution of recurrence relation.

#Q. Find the solution of the recurrence relation

$a_n = n\, a_{n-1}, \quad n \geq 1 \quad$ where $a_0 = 1$

$a_0 = 1$

$a_1 = 1 \cdot a_0 = 1 \cdot 1$

$a_2 = 2 \cdot a_1 = 2 \cdot 1 \cdot 1$

$a_3 = 3 \cdot a_2 = 3 \cdot 2 \cdot 1 \cdot 1$

$a_4 = 4 \cdot a_3 = 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1$

$\vdots$

$a_n = n \cdot a_{n-1} = n \cdot (n-1) \ldots \ldots 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1 = n!$

#Q. Find the solution of the recurrence relation

$$a_n = a_{n-1} + 3^{n-1}, \quad \text{where } a_1 = 2$$

$a_1 = 2$

$a_2 = a_1 + 3^{2-1} = 2 + 3^1$

$a_3 = a_2 + 3^{3-1} = 2 + 3^1 + 3^2$

$a_4 = a_3 + 3^{4-1} = 2 + 3^1 + 3^2 + 3^3$

$\vdots$

$a_n = 2 + 3^1 + 3^2 + 3^3 + \cdots + 3^{n-1}$

$= 1 + (1 + 3^1 + 3^2 + 3^3 + \cdots + 3^{n-1}) = 1 + \left(\frac{1 \cdot (3^n - 1)}{3 - 1}\right)$

$= 1 + \frac{3^n - 1}{2}$

$= \frac{2 + 3^n - 1}{2}$

$= \boxed{\left(\frac{3^n + 1}{2}\right)}$ Ans

Consider the following GP

$$a + a\cdot r + a\cdot r^2 + \cdots + a\cdot r^{n-1}$$

No. of terms in GP $= n$

first term $= a$

Common ratio $= r$

Summation of GP $= \dfrac{a(r^n - 1)}{(r-1)}$

#Q. Find the solution of the recurrence relation

$a_n = a_{n-1} + (2n + 1)$, where $a_0 = 1$

$a_0 = 1$

$a_1 = a_0 + (2.1+1) \quad = 1 + 3$

$a_2 = a_1 + (2.2+1) \quad = 1 + 3 + 5$

$a_3 = a_2 + (2.3+1) \quad = 1 + 3 + 5 + 7$

$\vdots$

$a_n = \quad + (2.n+1) \quad = 1 + 3 + 5 + 7 + \cdots + (2n+1)$

$= (2.0+1) + (2.1+1) + (2.2+1) + \cdots + (2.n+1)$

- Summation of first 'n' odd natural no. $= (n)^2$

- $\therefore$ $1+3+5+7+ - - - - - + (2n+1)$ is summation of first $(n+1)$ odd natural numbers $= (n+1)^2$

Consider the following A.P.

$$a + (a+d) + (a+2d) + - - - - + (a+(n-1)d)$$

No. of terms $= n$, first term $= a$, Common difference $= d$

$$\text{Summation of AP} = \frac{n}{2}\left[2 \cdot a + (n-1)d\right] = \frac{n}{2}\left[a + (a+(n-1)d)\right] = \frac{n}{2}\left[\text{first term} + \text{last term}\right]$$

$$a_n = 1 + 3 + 5 + 7 + \cdots + (2n+1)$$

$$= (1+2\cdot 0) + (1+2\cdot 1) + (1+2\cdot 2) + \cdots + (1+2n)$$

0 to n

total 'n+1' term.

$\therefore$ Summation $= \dfrac{n+1}{2}\left[1 + (2n+1)\right]$

$$= \frac{(n+1)}{2}\left[2n+2\right]$$

$$= \frac{(n+1)}{\cancel{2}} \cdot \cancel{2}(n+1) = (n+1)^2$$

#Q. Find the solution of the recurrence relation

$$a_n = a_{n-1} + \frac{1}{n(n+1)}, \quad \text{where } a_0 = 1$$

Using the method of partial fraction

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{(n+1)}$$

$a_0 = 1$

$a_1 = a_0 + \frac{1}{1} - \frac{1}{2} = 1 + \frac{1}{1} - \frac{1}{2}$

$a_2 = a_1 + \frac{1}{2} - \frac{1}{3} = 1 + \frac{1}{1} - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \frac{1}{3}$

$a_3 = a_2 + \frac{1}{3} - \frac{1}{4} = 1 + \frac{1}{1} - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \frac{1}{4}$

$a_n = \left(1 + \frac{1}{1}\right) \cdots - \frac{1}{(n+1)} \Rightarrow 2 - \frac{1}{(n+1)} = \left(\frac{2n+1}{n+1}\right)$

2 mins Summary

- **Topic** Recurrence relation
- **Topic** Formation of recurrence relation
- **Topic** Solution of recurrence relation using substitution method

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THANK - YOU

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